What if the spatial dimensions are not circular in shape? Do these remarkable conclusions about minimum spatial extent in string theory still hold? No one knows for sure. The essential aspect of circular dimensions is that they permit the possibility of wound strings. As long as the spatial dimensions—regardless of the details of their shape—allow strings to wind around them, most of the conclusions we have drawn should still apply. But what if, say, two of the dimensions are in the shape of a sphere? In this case, strings cannot get "trapped" in a wound configuration, because they can always "slip off" much as a stretched rubber band can pop off a basketball. Does string theory nevertheless limit the size to which these dimensions can shrink?

Numerous investigations seem to show that the answer depends on whether a full spatial dimension is being shrunk (as in the examples in this chapter) or (as we shall encounter and explain in Chapters 11 and 13) an isolated "chunk" of space is collapsing. The general belief among string theorists is that, regardless of shape, there is a minimum limiting size, much as in the case of circular dimensions, so long as we are shrinking a full spatial dimension. Establishing this expectation is an important goal for further research because it has a direct impact on a number of aspects of string theory, including its implications for cosmology.

Mirror Symmetry

Through general relativity, Einstein forged a link between the physics of gravity and the geometry of spacetime. At first blush, string theory strengthens and broadens the link between physics and geometry, since the properties of vibrating strings—their mass and the force charges they carry—are largely determined by the properties of the curled-up component of space. We have just seen, though, that quantum geometry—the geometry-physics association in string theory—has some surprising twists. In general relativity, and in "conventional" geometry, a circle of radius R is different from one whose radius is 1/R, pure and simple; yet, in string theory they are physically indistinguishable. This leads us to be bold enough to go further and ask whether there might be geometrical forms of space that differ in more drastic ways—not just in overall size, but possibly also in shape—but that are nevertheless physically indistinguishable in string theory.